Difference between Active Directory Domain Services (AD DS), Azure Active Directory (Azure AD) and Azure Active Directory Domain Services (Azure AD DS)

To provide applications, services, or devices access to a central identity, there are three common ways to use Active Directory-based services in Azure. This choice in identity solutions gives you the flexibility to use the most appropriate directory for your organization's needs. For example, if you mostly manage cloud-only users that run mobile devices, it may not make sense to build and run your own Active Directory Domain Services (AD DS) identity solution. Instead, you could just use Azure Active Directory.

Although the three Active Directory-based identity solutions share a common name and technology, they're designed to provide services that meet different customer demands. At high level, these identity solutions and feature sets are:

Active Directory Domain Services (AD DS)

Enterprise-ready lightweight directory access protocol (LDAP) server that provides key features such as identity and authentication, computer object management, group policy, and trusts.

AD DS is a central component in many organizations with an on-premises IT environment, and provides core user account authentication and computer management features.

For more information, see Active Directory Domain Services overview in the Windows Server documentation.

The Active Directory Domain Services (AD DS), is the traditional on-premises version of domain services provided by AD. Organizations use AD DS to centrally manage all their resource objects, such as users, computers, printers, shared folders, groups, organizational units (OUs), etc. These objects are part of the Active Directory domain, which allows the administrators to securely manage them through Group Policies. Some of the key features offered by AD DS includes:

1. One-premises identity & authentication
2. User and computer management
3. Group Policies
4. Domain trusts

AD DS is managed by the organizations on-premises. The Enterprise Administrators are responsible for managing AD DS domain controllers, AD sites, trust relationships between the domains, Group Policies, backing up and restoring AD DS, etc.

NOTE: In this article, the terms traditional AD and traditional AD DS, refer to the on-premises deployment of Active Directory and Active Directory Domain Services.

Azure Active Directory (Azure AD)

Azure Active Directory (Azure AD) is the Azure solution for identity and access management. Azure AD is a multitenant, cloud-based directory and identity management service from Microsoft. It combines core directory services, application access management, and identity protection into a single solution.

Cloud-based identity and mobile device management that provides user account and authentication services for resources such as Office 365, the Azure portal, or SaaS applications. Azure AD can be synchronized with an on-premises AD DS environment to provide a single identity to users that works natively in the cloud.

Azure AD offers some of the same features in the cloud, as AD DS offers on-premises. However, just because they both have AD in their names, doesn't mean they are identical services. Azure AD is a cloud-based identity service that offers the following:

1. Cloud-based identification & authentication
2. User and computer management
3. Mobile Device Management (MDM)
4. Access to Software as a service (SaaS) application, Microsoft Azure portal, and Office 365 services

Because Azure AD is hosted and managed by Microsoft in the cloud, organizations don't have direct access to AD domain controllers the way they do in their on-premises environment. Microsoft exposes parts of the Azure AD to organizations through the web-based interface so they have enough control to run and customize the services, but Microsoft is responsible for managing the services and servers behind the scenes in its datacenters across the globe.

Azure Active Directory Domain Services (Azure AD DS)

Provides managed domain services with a subset of fully-compatible traditional AD DS features such as domain join, group policy, LDAP, and Kerberos / NTLM authentication.

Azure AD DS integrates with Azure AD, which itself can synchronize with an on-premises AD DS environment. This ability extends central identity use cases to traditional web applications that run in Azure as part of a lift-and-shift strategy.

The Azure AD DS is a managed AD DS service in the cloud. In other words, if you want the traditional AD DS running in the cloud, you can take advantage of the Azure AD DS service by running AD DS under Azure AD. This means that you will be able to use traditional AD DS features, such as Kerberos and NTLM authentication, Group Policies (which aren't supported in Azure AD), LDAP, etc.

For organizations who are interested in running traditional AD DS services in the cloud, Microsoft offers a couple of methods. You can either use a *managed domain* or a *self-managed domain*. Here's the difference.

Managed Domain

A *managed domain* is something that you will create in the cloud using AD DS and Microsoft will create and manage the associated resources as necessary.

Self-Managed Domain

A *self-managed domain* is an AD DS environment that you can create in the cloud using the traditional tools. For example, you will use Virtual Machines (VMs) to install the AD DS domain controllers, member servers, etc. This is a self-managed domain so you (not Microsoft) will be responsible for managing the domain just like you do in your on-premises environment.

With Azure AD DS, the core service components are deployed and maintained for you by Microsoft as a *managed* domain experience. You don't deploy, manage, patch, and secure the AD DS infrastructure for components like the VMs, Windows Server OS, or domain controllers (DCs).

Azure AD DS provides a smaller subset of features to traditional self-managed AD DS environment, which reduces some of the design and management complexity. For example, there's no AD forests, domain, sites, and replication links to design and maintain. For applications and services that run in the cloud and need access to traditional authentication mechanisms such as Kerberos or NTLM, Azure AD DS provides a managed domain experience with the minimal amount of administrative overhead.

When you deploy and run a self-managed AD DS environment, you have to maintain all of the associated infrastructure and directory components. There's additional maintenance overhead with a self-managed AD DS environment, but you're then able to do additional tasks such as extend the schema or create forest trusts.

Azure AD DS and Azure AD

Azure AD lets you manage the identity of devices used by the organization and control access to corporate resources from those devices. Users can also register their personal device (a bring-your-own (BYO) model) with Azure AD, which provides the device with an identity. Azure AD then authenticates the device when a user signs in to Azure AD and uses the device to access secured resources. The device can be managed using Mobile Device Management (MDM) software like Microsoft Intune. This management ability lets you restrict access to sensitive resources to managed and policy-compliant devices.

Traditional computers and laptops can also join to Azure AD. This mechanism offers the same benefits of registering a personal device with Azure AD, such as to allow users to sign in to the device using their corporate credentials.

Azure AD joined devices give you the following benefits:

- Single-sign-on (SSO) to applications secured by Azure AD.
- Enterprise policy-compliant roaming of user settings across devices.
- Access to the Windows Store for Business using corporate credentials.
- Windows Hello for Business.
- Restricted access to apps and resources from devices compliant with corporate policy.

On an Azure AD-joined or registered device, user authentication happens using modern OAuth / OpenID Connect based protocols. These protocols are designed to work over the internet, so are great for mobile scenarios where users access corporate resources from anywhere.

With Azure AD DS-joined devices, applications can use the Kerberos and NTLM protocols for authentication, so can support legacy applications migrated to run on Azure VMs as part of a lift-and-shift strategy

Aspect	**Azure AD-joined**	**Azure AD DS-joined**
Device controlled by	Azure AD	Azure AD DS managed domain
Representation in the directory	Device objects in the Azure AD directory	Computer objects in the Azure AD DS managed domain
Authentication	OAuth / OpenID Connect based protocols	Kerberos and NTLM protocols
Management	Mobile Device Management (MDM) software like Intune	Group Policy
Networking	Works over the internet	Must be connected to, or peered with, the virtual network where the managed domain is deployed
Great for…	End-user mobile or desktop devices	Server VMs deployed in Azure

Feature	**Azure AD DS**	**Self-managed AD DS**
Managed service	✓	✕
Secure deployments	✓	Administrator secures the deployment
DNS server	✓ (managed service)	✓
Domain or Enterprise administrator privileges	✕	✓
Domain join	✓	✓
Domain authentication using NTLM and Kerberos	✓	✓
Kerberos constrained delegation	Resource-based	Resource-based & account-based

Custom OU structure	✓	✓
Group Policy	✓	✓
Schema extensions	✕	✓
AD domain / forest trusts	✓ (one-way outbound forest trusts only)	✓
Secure LDAP (LDAPS)	✓	✓
LDAP read	✓	✓
LDAP write	✓ (within the managed domain)	✓
Geo-distributed deployments	✕	✓

Ref: https://ramprasadtech.com/difference-between-azure-ad-azure-ad-ds-and-ad-ds/